Power Transients in Optical-Hybrid Amplifiers: Theoretical and Simulation Analysis in a Real-Case Scenario of the NSFNet

Bárbara Dumas Feris and Ricardo Olivares

***Abstract*—This paper focuses on Wavelength-Routed-Optical Networks (WRONs) with a variable load (number of propagated channels) using optical Hybrid Amplifiers (HAs). In this context, we take a real-case scenario of the highest-load-variation route of the NSFNet covering 5580 km, we redesign one of its links with a cascade of 13 optical HAs (Erbium-Doped-Fiber Amplifier (EDFA) + backward-pumped Distributed-Fiber-Raman Amplifier (DFRA)) for C-band-optical transmission and we assess their power gain under the network dynamics. When increasing the number of propagated Wavelength-Division Multiplexing (WDM) channels from 6 to 17, HAs experience critical transient and steady-state power excursions. The effective adaptation and application of the link-control technique allow us to reduce these power excursions from 4.61 dB to 0.73 dB and from 4.07 dB to 0.14 dB for transient and steady-state regimes, respectively. This technique proved to be simple and effective, presenting a high potential to be tested in other wavelength bands.**

***Index Terms*—DFRAs, EDFAs, HAs, power-transient response, WRONs.**

I. Introduction

OPTICAL Hybrid Amplifiers (HAs) are defined as the integration of rare-earth Doped-Fiber Amplifiers (DFAs) and Distributed-Fiber-Raman Amplifiers (DFRAs) as a whole configuration [1], [2], [3], [4], [5], [6], [7]. In the early 2000s, one of the main research goals was to extend the gain bandwidth of the already installed networks using Erbium-Doped-Fiber Amplifiers (EDFAs) beyond the C-band-optical transmission [1], [2], [3]. This was typically achieved by installing Raman pumps in strategically selected nodes—either end or intermediate nodes—depending on the span length and the availability of electrical and physical infrastructure [4], [5]. This not only allowed to exploit the main advantages of both amplifiers, but also represented (and still does) an attractive economic approach. Among these advantages, we find for EDFAs: high gain and high power-conversion efficiency; for DFRAs: low noise, a distributed gain that reduces fiber nonlinearities and a gain-bandwidth design that can be adjusted in terms of spectral location, width and flatness according to the Raman-pumping scheme [7].

Later on, the research community has been focused on facing the capacity crunch [8], [9]. Three main approaches have been proposed: 1. to extend the transmission to multiple wavelength bands (E, S, C, L) [10], [11]; 2. the development of Space-Division Multiplexed (SDM) technologies [12], [13] (by the use of Multi-Core Fiber (MCF) [14] or Few-Modes Fiber (FMF) [15] or even both at the same time [16]); and 3. the increase of the efficiency of resource allocation by optimization strategies [17].

Nowadays, HAs are key devices to address the first approach and have been experimentally proven in multi-band environments [18], [19], [20], [21]. E.g. the transmission of 301 Tb/s over several optical bands (E, S, C, L) has been demonstrated in [21].

In this paper, we focus on C-band-optical HAs. However, it should be noticed that the principle can be easily extended to multi-band-optical HAs. Particularly, we address the challenges imposed by Wavelength-Routed-Optical Networks (WRONs) where the overall HA-input power can dynamically change according to the network needs, leading to a degradation of the network performance. Actually, we must be able to guarantee the propagation of a variable load (number of channels) when adding/removing channels at intermediate nodes, due to a dynamic reconfiguration of the network in case of link failure or due to an increase of the network's capacity.

The fact of dealing with a variable load leads to power transients (or power-gain excursions) that can degrade the network's performance. On one hand, when channels are dropped, the power of the remaining channels may exceed the threshold where fiber nonlinearities cannot be ignored. On the other hand, when channels are added, the power of the surviving channels shall decrease and may fall below the receiver sensitivity. Both scenarios can significantly damage the Optical-Signal-to-Noise Ratio (OSNR).

In order to maintain the overall network's performance, the power of the surviving channels must be kept within acceptable limits. For that, several transient-control techniques have been applied to both EDFAs [22], [23], [24] and DFRAs [25], [26], separately. Although the transient response in cascades of EDFAs [27], [28] and DFRAs [29], [30] has been extensively

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studied, the transient response of HAs has received limited attention [6], [31], [32]. From what we can gather, in addition to our previous work [33], no studies have been reported on the cumulative effects of the transient response in cascades of HAs and their control. More, there is no assessment of this problem neither of its control in a real-case scenario. Therefore, in this work, for the first time to our knowledge, we investigate, through numerical simulation, the transient response of a cascade of HAs along a link of the NSFNet that experiences high-load variations. We then apply the link-control technique to mitigate the unacceptable power excursions that result from these variations. By simulating a real-world scenario, this study provides valuable insights, empowering network operators to implement proactive strategies for ensuring a consistent and reliable quality of service.

Hence, in Section II we describe the amplification configuration used for building a cascade of HAs to compensate losses of an entire link of the NSFNet. We also give detailed information about the link-control module used for reducing power excursions. Section III describes the NSFNet and the real-case scenario where we study power transients and their mitigation. Section IV shows simulation results in terms of optical power and power excursions over time when the load variations occur. We give a detailed analysis about the behavior of HAs over the cascade and also the impact of using the link-control technique. Finally, Section V draws the conclusion.

II. Power Transients in Cascades of Hybrid Amplifiers

A. Hybrid-Amplifier Design

The design of a single stage of hybrid amplification that compensates attenuation of 150 km of Standard-Single-Mode Fiber (SSMF) has been thoroughly studied under four different configurations in [31]. For two of these configurations the EDFA is located first, followed by the DFRA (EDFA + DFRA). The other two configurations invert the order of the amplifiers (DFRA + EDFA). For both combinations, backward- and forward- DFRA-pumping schemes are considered. Regardless of the configuration, the design of each amplifier of a single stage of HA goes as follows: on one hand, the EDFA compensates (as a post-amplifier) 100 km of SSMF-power losses, which are estimated to be 20 dB. The EDFA consists of a 12 m long Erbium-Doped Fiber (EDF) forward pumped, without any gain-control scheme. On the other hand, the DFRA compensates 50 km of SSMF-power losses (10 dB), thanks to a flat-gain algorithm for HAs [7], which outcome allows a ripple of ≈ 1 dB (before load variation). Simulation parameters of EDFAs and DFRAs are specified in [34] and [30], respectively. Fig. 1 presents one of these hybrid configurations: "EDFA + backward DFRA", composed of an EDFA followed by a backward-pumped DFRA.

In our previous work [31], [32], [33], we have developed a simulation tool in order to analyze power transients in HAs, in cascades of 12 HAs and by applying the link-control technique with the aim of mitigating undesirable-power excursions under high-load-variation scenarios, for the configurations described above. The theoretical model used for the EDFA in time domain [35] neglects the Amplified-Spontaneous-Emission (ASE) noise; since, while working at high-saturation levels this noise is negligible compared to the signal power. The theoretical model used for the DFRA [25] considers effects of attenuation, ASE noise, Rayleigh scattering, stimulated-Raman scattering, anti-Stokes- and Stokes- spontaneous scatterings. When using a small channel spacing ($<$ 1nm), the anti-Stokes process cannot be neglected compared to the Stokes one [30]. The propagation equation of this model describes power variation of signals and pumps over position and time. Both models are solved using numerical integration techniques thanks to a C-programming code of our own performed on a personal computer. The fourth-order-Runge–Kutta method is considered to solve the EDFA model. The finite-difference method in combination with the relaxation method (for backward-pumping schemes) is applied to solve the DFRA model.

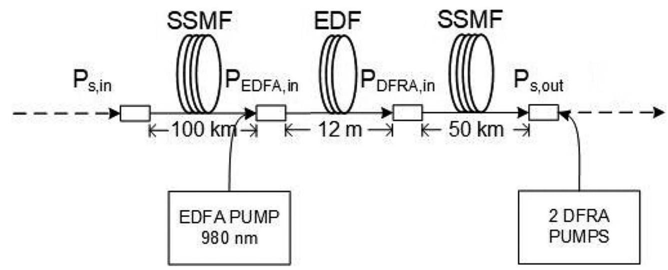

Fig. 1. Single stage of hybrid amplification under "EDFA + backward DFRA" configuration (150 km). $P_{s,in}$: signal-input power; $P_{EDFA,in}$: EDFA -input power; $P_{DFRA,in}$: DFRA-input power; $P_{s,out}$: signal-output power.

According to our analysis in [31], at high-load variations, the use of backward-pumped DFRAs into HAs limits power excursions under transient and steady-state regimes. This behavior has been identified not only in one single HA, but also in cascades of HAs. Moreover, the link-control technique is more effective in reducing such power excursions when using backward-pumped DFRAs [32], [33]. In addition, slight differences regarding the order of the amplifiers inside the HA have been reported in [31], [32], [33]. The fact of placing the backward-pumped DFRA after the EDFA (EDFA + DFRA) makes the mitigated power excursions smoother, i.e. shorter and slower overshoots and shorter steady-state levels compared to the configuration that inverts the order of the amplifiers (DFRA + EDFA). As a consequence, in this paper we will focus our study on what we consider the best HA configuration: "EDFA + backward DFRA" (Fig. 1).

B. Link-Control Technique

The link-control technique aims to reduce power-transient effects by keeping the amplifier's load constant whenever there are changes in the number of propagated-WDM channels, due to the reconfiguration of the network. This technique was originally designed for WDM networks using EDFAs [22], it was then studied by computer simulation for WDM networks using DFRAs [30] and HAs [33]. In this paper, we apply the link-control technique to cascades of HAs in a real-case scenario with a critical-load variation. Fig. 2 shows the link-control

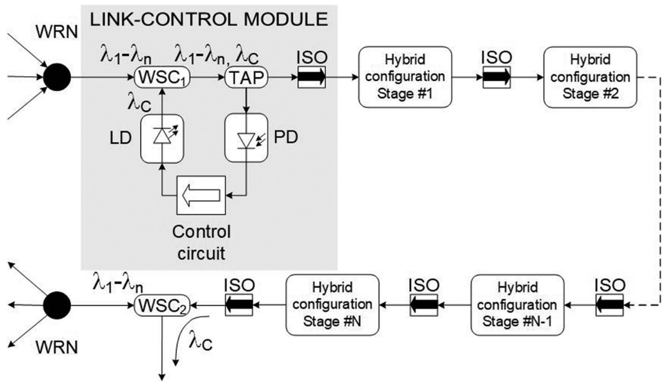

Fig. 2. Scheme of the link-control module in a cascade of HAs.

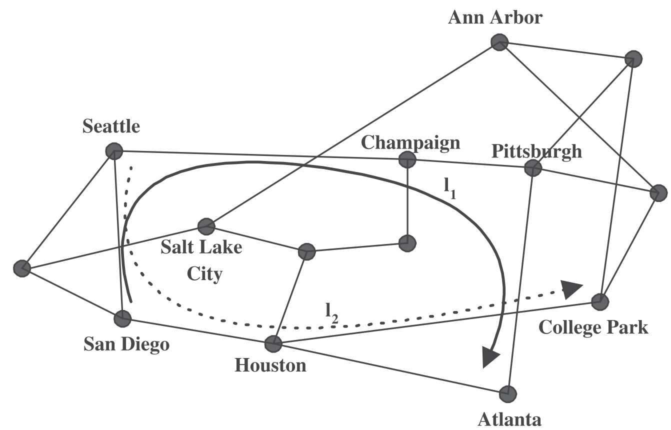

Fig. 3. Network-interconnection diagram of NSFNet [34].

module considered in our study. It consists in a control channel at wavelength λ_C (inside the transmission band), generated by the Laser Diode (LD). The control channel is inserted to the link through a Wavelength-Select Coupler (WSC), WSC$_1$, before the first hybrid-amplification stage of the cascade. This channel is removed after the last amplification stage of the cascade through WSC$_2$ in order to avoid unnecessary load in the network. The input power of λ_C is determined by a Proportional-Integrative (PI) control module [36] that takes as reference the total-input power detected by a Photo-Detector (PD), so that the total power at the input of the first stage of hybrid amplification is kept constant. This effect is transmitted along the cascade, therefore, undesirable-power excursions in the transient response can be reduced. Isolators (ISOs) are used between every amplification stage to avoid pump transfers to the control module.

The results presented in this paper are generated by means of computer simulation. This means that the reference taken into account by the link-control module (of Fig. 2) is the total-input power calculated by solving numerically the theoretical models mentioned above (which is equivalent of using the sample taken by the PD as a reference).

The input power of the control channel, $P_{cc}(t)$, regarding time t is calculated according to (1). This equation describes the behavior of the PI control implemented inside the feedback loop.

$$\frac{dP_{cc}(t)}{dt} = \frac{k_p}{T_r}R(t) + k_p T_r \frac{dR(t)}{dt} \tag{1}$$

Where k_p and T_r represent the proportional gain and the integrative reset time, respectively. $R(t)$ corresponds to the error function regarding time t as defined in (2),

$$R(t) = |P_T(t=0) - P_T(t)| \tag{2}$$

where

$$P_T(t) = \sum_i P_i(t) \tag{3}$$

is the total-input power, that means the summation of every WDM-channel-input power at wavelength λ_i (including the control channel at wavelength λ_c) as a function of time t.

k_p and T_r were set considering a feedback-loop-response time of 4 μs (typically associated to real-electronic components [30]) and the Ziegler-Nichols method [36], resulting $k_p = 1$ and $T_r = 1$ μs.

Even if the PI control ensures a steady-state error equal to zero, it does not guarantee that steady-state-power excursions will be null for the surviving channels. The power gain associated to a signal does not depend exclusively on the amplifier-saturation level, but on its wavelength, on its input power and on the other channels wavelength and input power. For this reason, the control channel must be chosen at a wavelength such as the steady-state-power gain corresponds to the average-power gain of all the transported channels.

III. Real-Case Simulation: Application to the NSFNet

A. Description of NSFNet

The NSFNet consists of 14 nodes interconnected through 21 bi-directional-fiber links [34], [37], [38]. It interconnects several research centers in the United States of America. These nodes can be terminals or Wavelength-Routed Nodes (WRNs). The network-interconnection diagram is depicted in Fig. 3. l_1 corresponds to the longest path; while, l_2 is the path with the highest variation in the number of propagated channels (load variation) whenever the network presents a link failure.

B. Simulation Scenario and Amplification Parameters

We focus our study on path l_2, representing a critical-real-case scenario. This path connects the following nodes: Seattle, San Diego, Houston and College Park, covering 5580 km. Under normal-operating mode each link carries 6, 6 and 13 channels, respectively. However, under restoration-operating mode—failure of the Salt Lake City-Ann Arbor link—this number changes to 6, 17 and 23, respectively. Here we study the San Diego-Houston link, when it goes from normal- to restoration-operating mode, going from 6 to 17 channels.

This network considers the propagation of 23 WDM channels in the range 1545.0 - 1558.2 nm, evenly spaced at 0.6 nm. Channels are numbered from #1 to #23, where channel #n is

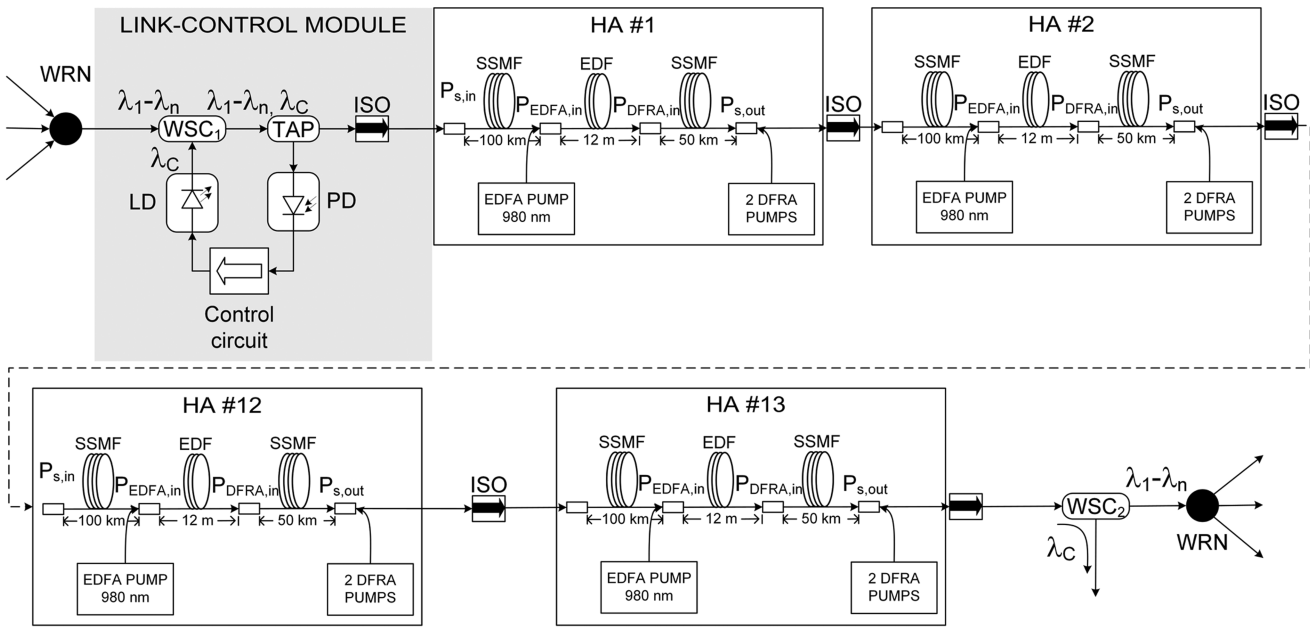

Fig. 4. Full simulation schematic of a cascade of 13 HAs (1950 km), under "EDFA + backward DFRA" configuration, using the link-control module to reduce transient-power excursions. $P_{s,in}$: signal-input power; $P_{EDFA,in}$: EDFA -input power; $P_{DFRA,in}$: DFRA-input power; $P_{s,out}$: signal-output power.

located at $\lambda_n = 1545.0 + (n-1) \cdot 0.6$ nm. For the particular case of the San Diego-Houston link, only 6 channels (# 1, 2, 3, 6, 8 and 10) are propagated from time $t = 0$ ms; then, at time $t = 1$ ms, 11 additional channels are inserted (# 4, 5, 7, 9, 11, 12, 13, 15, 17, 18 and 21), the same 11 channels are next removed at time $t = 2$ ms. Note that the newly added/removed channels are simulated as step functions (off-on/on-off).

We introduce HAs to the San Diego-Houston link (covering 1920 km). Our design is based on the hybrid configuration "EDFA + backward DFRA", because it has proven to be the best option to reduce power transient under high-load-variation scenarios (refer to Section II-A). We consider a signal-input power of $P_{s,in} = 0$ dBm/channel, aiming to get saturated optical amplifiers even at low load. Each HA is designed to compensate the SSMF-power losses over 150 km. Therefore, a cascade of 13 HAs is required to cover a distance of 1920 km. Pump powers are set to: $P_{p,EDFA} = 30$ mW at wavelength $\lambda_{p,EDFA} = 980$ nm, for the forward-pumped EDFA; $P_{p1,DFRA} = 110$ mW at wavelength $\lambda_{p1,DFRA} = 1455$ nm and $P_{p2,DFRA} = 60$ mW at wavelength $\lambda_{p2,DFRA} = 1463$ nm, for the backward-pumped DFRA. At each amplification stage, after the first 100 km of SSMF, we reach an EDFA-input power of $P_{EDFA,in} \approx -20$ dBm/channel and an EDFA-output power of $P_{EDFA,out} \approx 0$ dBm/channel. Notice that the DFRA-input power is equal to the EDFA-output power. Along the last 50 km under Raman amplification, 10 dB of SSMF losses are distributively compensated, so the signal-output power reaches $P_{s,out} \approx 0$ dBm/channel.

The control channel was placed within the spectral band of the signal channels to achieve a comparable power gain. Its initial power matches that of the surviving channels before the add/drop action, maximizing the dynamic range for effective load compensation. This configuration enables a rapid response, with a rise time in the microseconds range. We look for a wavelength with a steady-state gain as close as possible to the average steady-state gain of all channels. Recall that, even if our configuration presents low ripple (≈ 1 dB), the gain is slightly different for each wavelength. This criterion avoids changing the shape of the spectral gain after the load variation. It turns out that channel # 14 ($\lambda_C = \lambda_{14} = 1552.8$ nm) has the closest match. Fig. 4 presents the full schematic of simulation through a cascade of 13 HAs using the link-control module to reduce transient-power excursions.

IV. Simulation Results

The transient response of HAs, under the scenario described above, is analyzed using three key metrics: steady-state power (P), defined as a stable final value reached after a settling time [36]; overshoot ($P_{\max}$), representing the peak power by which the response exceeds the steady-state level; and undershoot ($P_{\min}$), referring to the minimum power by which the response temporarily falls below the steady-state level. Fig. 5 shows power of channels 9 and 10 as a function of time, through a cascade of 13 HAs under the configuration "EDFA + backward DFRA" for stages # 1, 7 and 13. Transient (overshoots $P_{\max}$ and undershoots $P_{\min}$) and steady-state (P) values of both channels are summarized in Table I. These results reveal the absence of any per-amplifier control of gain.

Channel 9 ($\lambda_9 = 1549.8$ nm) is one of the new channels added at $t = 1$ ms and channel 10 ($\lambda_{10} = 1550.4$ nm) is one of the channels propagated since $t = 0$ ms. It can be seen that, during 1 ms $< t <$ 2 ms, power of channel 9 (continuous curves) reaches transient and steady-state values below those of channel

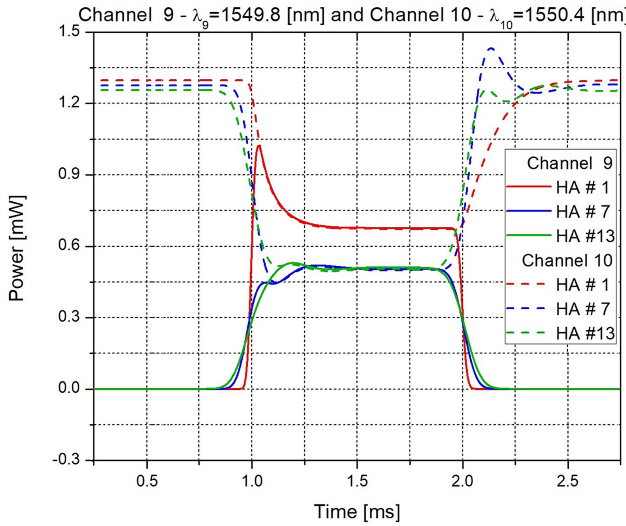

Fig. 5. Optical power of channel 9 ($\lambda_9 = 1549.8$ nm) and channel 10 ($\lambda_{10} = 1550.4$ nm) as a function of time, through a cascade of 13 HAs under "EDFA + backward DFRA" configuration, for stages # 1, 7 and 13. When 11 out from 17 WDM channels are added at time $t = 1$ ms and then dropped at time $t = 2$ ms, while 6 other channels are already in propagation.

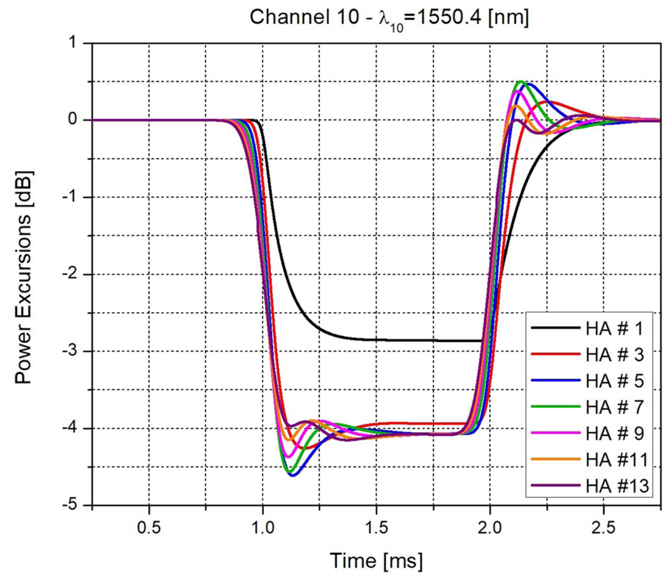

Fig. 6. Optical power excursion of channel 10 ($\lambda_{10} = 1550.4$ nm) as a function of time, through a cascade of 13 HAs under "EDFA + backward DFRA" configuration, for the odd HA stages. When 11 WDM channels are added/dropped, while 6 other channels are already in propagation.

TABLE I
OPTICAL POWER: TRANSIENT AND STEADY-STATE VALUES

HA	Channel 9 $\lambda_9 = 1549.8$ nm		Channel 10 $\lambda_{10} = 1550.4$ nm	
#	P_{max} [mW]	P [mW]	P_{min} [mW]	P [mW]
1	1.013	0.675	0.675	0.675
7	0.523	0.521	0.524	0.522
13	0.525	0.522	0.521	0.521

P_{max}: power overshoots, P_{min} power undershoots and P: steady-state power.

TABLE II
SIMULATION RESULTS THROUGH A CASCADE OF 13 HAS (CHANNEL 10)

HA	Channel 10, $\lambda_{10} = 1550.4$ nm			
#	ΔP_{max} $[dB]$	ΔP $[dB]$	t_R $[\mu s]$	t_s $[\mu s]$
1	−2.86	−2.86	190	826
3	−4.25	−3.94	113	920
5	−4.61	−4.07	107	959
7	−4.56	−4.07	115	960
9	−4.37	−4.07	123	958
11	−4.14	−4.07	130	955
13	−4.07	−4.07	144	953

ΔP_{max}: power-excursion peak, ΔP: steady-state-power excursion, t_R: rise time and t_s: settling time.

10 before $t = 1$ ms, as expected. Power of channel 10 (dashed curves) decreases at $t = 1$ ms and reaches its original value after $t = 2$ ms. For channel 9 power overshoots are observed at $t = 1$ ms, they become smaller for stages # 7 and 13. For channel 10 instead, power undershoots and overshoots can be seen at $t = 1$ ms and $t = 2$, respectively. They get their largest absolute value at stage #7. For both channels steady-state-power values are the same at each stage of HA. The latter increases (decreases) for channel 9 (10) from stage #1 to stage #7 and keeps the same until stage #13.

We explain this behavior as follows, prior to the addition of new channels, the power gain delivered by the amplifiers corresponds to that stimulated by 6 channels. For EDFAs, this depends on the availability of erbium ions at the metastable level; for DFRAs, on the available carriers for the stimulated-Raman scattering to be produced; both cases depend on the saturation level given by both pumps and input power. Notice that both amplifiers are already saturated with a load of 6 channels. As a consequence, immediately after adding channels, the new channels compete with the existing ones, sharing the remaining population inversion among all 17 channels. This increased competition triggers a transient response, characterized by a reduction in the gain of the existing channels (undershoots). The system does not immediately reach a new steady state: there is an exponential recovery as the population inversion reaches a new balance, i.e. an equilibrium between absorption and emission rates (depending on pump dynamics of both amplifiers). Under this new condition, the per-channel-power gain is lower than the previous one, due to a stronger saturation of the amplifiers.

After the new channels are dropped, the opposite effect occurs, that is to say, overshoots and higher steady-state values (restoring the initial condition) on the surviving channels. After stage #1, the input power of the surviving channels is constantly modified by the transient effects introduced by the previous stages, changing the saturation level of the next stage. For channel 9 (at $t = 1$) an overshoot means more stimulation to the HA, meaning more gain (higher overshoots for next stages) unless the latter gets even more saturated, meaning less gain (lower overshoots for next stages). For channel 10, we can see the opposite effect. For a better understanding of these dynamics, we introduce Fig. 6, showing power excursions of channel 10 ($\lambda_{10} = 1550.4$ nm) as a function of time, through a cascade of 13 HAs under "EDFA + backward DFRA" configuration, for the odd HA stages. Table II summarizes these results in terms

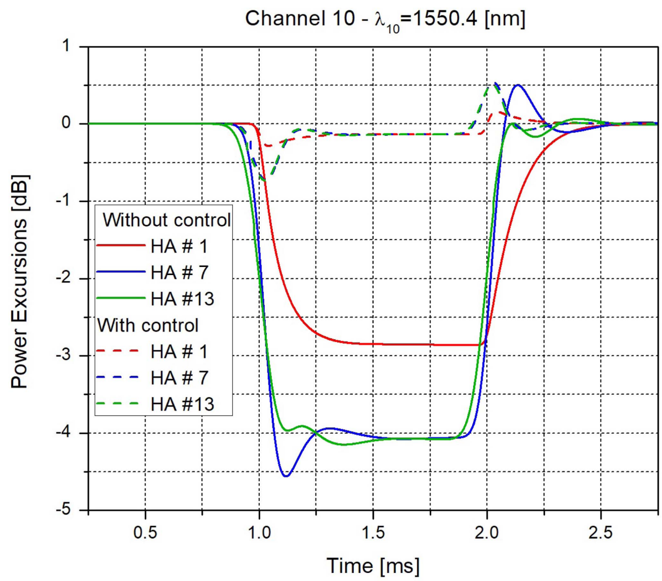

Fig. 7. Optical power excursion of channel 10 ($\lambda_{10} = 1550.4$ nm) as a function of time, through a cascade of 13 HAs under "EDFA + backward DFRA" configuration, for the HA stages # 1, 7 and 13. When 11 WDM channels are added/dropped, while 6 other channels are already in propagation. The link control technique is applied by using the compensation channel (14) at wavelength $\lambda_{14} = 1552.8$ nm.

TABLE III
SIMULATION RESULTS THROUGH A CASCADE OF 13 HAS WHEN USING THE LINK-CONTROL TECHNIQUE (CHANNEL 10)

HA	Channel 10, $\lambda_{10} = 1550.4$ nm			
#	ΔP_{max} $[dB]$	ΔP $[dB]$	t_R $[\mu s]$	t_s $[\mu s]$
1	-0.28	-0.13	38	510
7	-0.73	-0.14	75	903
13	-0.73	-0.14	75	935

ΔP_{max}: power-excursion peak, ΔP: steady-state-power excursion, t_R: rise time and t_s: settling time.

of: power-excursion peak $\Delta P_{\max}$ (in dB); steady-state-power excursion ΔP (in dB); rise time t_R (in μs), time from 10% to 90% of steady-state level; and settling time t_s (in μs), required time to reach steady state.

It can be seen that the power excursions experienced by channel 10 are negative; this means that the power of this channel decreases after the insertion of the new channels and that it recovers its original value when those channels are further removed. The evolution of power excursions over the odd amplification stages allows us to locate the inflection point where the absolute value of $\Delta P_{\max}$ starts to decrease: after stage # 5. Here, the HA is more saturated and does not allow to deliver more gain to the new added channels, so both rise and settling times increase.

We now apply the link-control technique to the same case scenario. Fig. 7 shows the power excursions experienced by channel 10 as a function of time, without any kind of control under continuous curves and with link control under dashed curves, for HA stages # 1, 7 and 13. These results are summarized in Table III in terms of: power-excursion peak $\Delta P_{\max}$ (in dB), steady-state-power excursion ΔP (in dB), rise time t_R (in μs) and settling time t_s (in μs).

It can be noticed that power excursions of channel 10 when applying the link-control technique (dashed curves) present undershoots and overshoots just after adding and dropping channels. We observe a behavior similar to undershoots and overshoots presented when no control technique is applied (continuous curves), but with smaller absolute values that also are amplified through the cascade of HAs. For instance, according to Tables II and III, undershoots values ($\Delta P_{\max}$) after $t = 1$ ms can go from -2.86 dB to -0.28 dB, for HA stage #1; and from -4.07 dB to -0.73 dB, for HA stage #13. These undershoots become faster when using the link-control technique, eg. rise times (t_R) can be reduced by half for HA stage #13. In addition, we can observe that steady-state-power excursions are reduced at different ratios according to the HA stage. For example, for HA stage #1 the steady-state power excursions of channel 10 go from -2.86 dB to -0.13 dB; instead, for the HA stages # 7 and 13 they go from -4.07 dB to -0.14 dB. This can be explained since the steady-state-power variations, despite being small, accumulate through the cascade, slightly modifying the saturation of the HA. This means that the HA will increase power-excursions of the next stage, repeating the cycle for further stages. Meanwhile, settling times (t_s) can only be reduced for HA stage #1, for the other stages it keeps more or less the same.

We now want to see what happens with the other channels, hence, Fig. 8 presents steady-state optical power as a function of wavelength (propagated 17 WDM channels) before (black squares) and after (red circles without control and yellow circles with control) adding 11 WDM channels. Results without control are presented on top; results when applying the link-control technique (using a compensation channel at a wavelength $\lambda_{14} =$ 1552.28 nm) are presented at the bottom. These results are shown for HA stage #1 (Fig. 8(a)) and for HA stage #13 (Fig. 8(b)).

The behavior (previously observed) of channel 10 is reproduced by the other surviving channels (other channels propagated from $t = 0$ ms, as channel 10) for all cases of Fig. 8. As shown in Fig. 8(a) (top), the average power after adding channels (red circles) is lower than before adding them (black squares) in ≈ 2.5 dB. Under this new condition, the channels not only reach a lower gain, but also present a higher ripple (≈ 3.5 dB). The differences observed between each channel are justified according to the HA-gain spectrum, which has a peak at $\lambda_{10} = 1550.4$ nm (channel 10). Once the control technique is applied, Fig. 8(a) (bottom), the surviving channels keep almost the same power as before adding new ones. Moreover, the ripple can be reduced by 1 dB (yellow circles). Note that channel 14 has absorbed the gain differences and now reaches a power almost 10 dB lower than before adding channels.

A similar behavior is observed for stage #13, with slight differences compared to stage #1: 1. when no control is applied, Fig. 8(b) (top), the average power after adding channels (red circles) is much lower than before adding them (black squares) in ≈ 5 dB; 2. when the control is applied, Fig. 8(b) (bottom), power values after adding channels (yellow circles) still remain very close to the references (black squares).

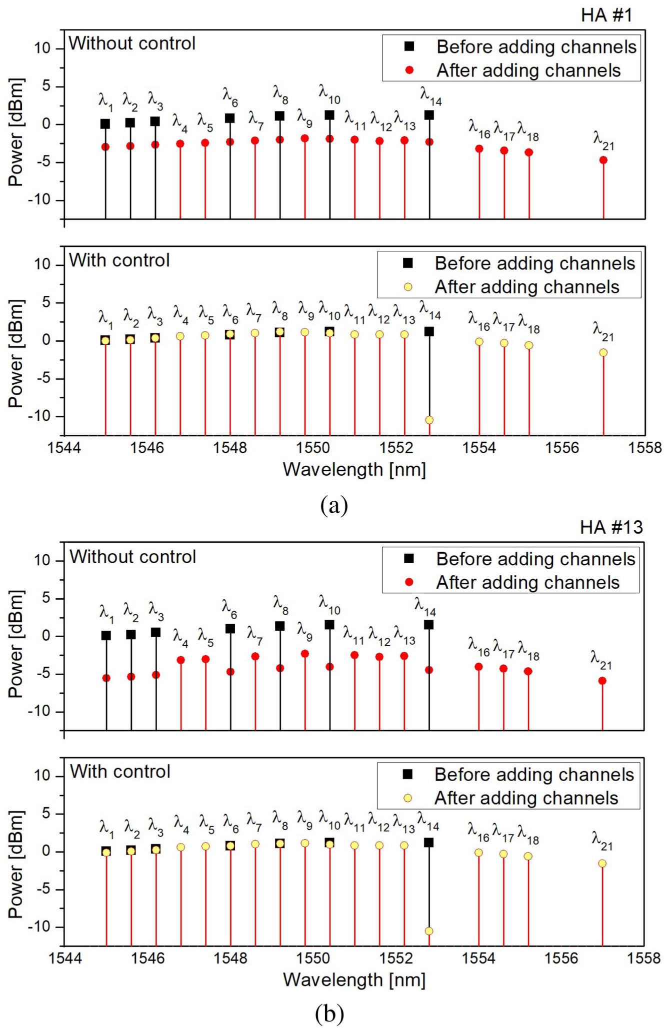

Fig. 8. Steady-state optical power as a function of wavelength (17 propagated WDM channels) before and after adding 11 WDM channels; for HA stages #1 (a) and #13 (b); without (on top) and with (at the bottom) applying the link-control technique.

The principle keeps the same, even if there is an important reduction of undesirable transient and steady-state power excursions by means of the compensation channel, steady-state-power excursions, despite of reaching low values, accumulate through the cascade. This effect induces slight variations on the saturation of HAs, which in turn increases the power excursions of the next stage, modifying the gain-spectrum characteristic at each stage, repeating the cycle for the forward stages of hybrid amplification.

V. Conclusion

In this paper, we analyze a realistic case of adding and dropping channels in a WRON using optical HAs. More specifically, we study one of the links of the highest-load-variation route of the NSFNet, that covers 5580 km. For this link, we design an amplification scheme with a cascade of 13 HAs under an "EDFA + backward DFRA" configuration. We study the transient and steady-state responses of the amplifiers when the load changes from 6 to 17 propagated channels by computer simulation. We describe and analyze these behaviors by calculating the output power and the power excursions of the surviving channels and the recently added channels at each stage of hybrid amplification. These load variations lead to changes in the saturation level of the amplifiers modifying the per-channel gain. These undesirable-power excursions reached maximum values of: 4.61 dB for the transient response (overshoot) and 4.07 dB for the steady-state response, both after the fifth stage of hybrid amplification.

We then studied the link-control technique, we set the PI-controller parameters and adjusted the spectral location of the compensation channel in order to maximize the control effect, trying to get a uniform gain among all channels. This technique proved to be simple and effective: it allowed to successfully reduce power excursions to 0.73 dB and 0.14 dB for transient and steady-state responses, respectively.

Both the power-transient analysis and the implementation of the link-control technique applied to real-case scenario, are to be assessed in multiband-optical networks using optical-hybrid amplifiers (composed by other kinds of DFAs + DFRAs). Even if some different physical phenomena are to be expected, the main behavior of variation on the saturation level of the amplifiers will govern the per-channel gain. Hence, the link-control technique shall be easily adapted and applied to the other transmission bands.

Finally, as these results were generated by means of computer simulation, our future work considers their experimental validation, mainly to assess the link-control technique with real optical devices.

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